

① Processos Estocásticos (Dia 23/09/2026)

Última aula:

Teo (1º Teo da Parada Opcional): $(M_n)_{n \in \mathbb{N}}$ martingal, τ ~~t. de p.~~ t.p. finito. $(M_{n \wedge \tau})_{n \in \mathbb{N}}$ limitado (isto é, $\exists C \geq 0: |M_{n \wedge \tau}| \leq C \forall n$)

$$\Rightarrow \mathbb{E}(M_\tau) = \mathbb{E}(M_0)$$

prova: ~~$\tau < \infty$~~ $\forall \omega \in \{\tau < \infty\}$, ~~$M_{n \wedge \tau}$~~ $\forall n \geq \tau(\omega)$ vale

$$M_{\tau \wedge n}(\omega) = M_\tau(\omega)$$

$$\Rightarrow \forall \omega \in \{\tau < \infty\}, M_{\tau \wedge n}(\omega) \xrightarrow{n \rightarrow \infty} M_\tau(\omega)$$

$$\Rightarrow \mathbb{P}(\tau < \infty) = 1. \text{ Logo } M_{\tau \wedge n} \xrightarrow{n \rightarrow \infty} M_\tau(\omega) \text{ q.c.}$$

$$\begin{aligned} \bullet |M_{\tau \wedge n}| \leq C \quad \forall n \in \mathbb{N} &\xRightarrow{\text{TCD}} \mathbb{E}(M_\tau) = \mathbb{E}\left(\lim_{n \rightarrow \infty} M_{\tau \wedge n}\right) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(M_{\tau \wedge n}) = \lim_{n \rightarrow \infty} \mathbb{E}(M_0) \\ &= \mathbb{E}(M_0) \end{aligned}$$

Teo (Parada Opcional de Doob): $(M_n)_{n \in \mathbb{N}}$ martingal, τ t.p.

$\Rightarrow \mathbb{E}(M_\tau) = \mathbb{E}(M_0)$ se valer algum item abaixo:

(i) ~~$\exists C \geq 0$ t.q. $|M_n| \leq C \forall n$~~ τ é limitado ($\tau \leq C$ q.c.)

(ii) $(M_{\tau \wedge n})_{n \in \mathbb{N}}$ é limitado e τ é finito

(iii) $\mathbb{E}(\tau) < \infty$, $|M_k - M_{k-1}| \leq C$, pra algum $C > 0$

(iv) $\mathbb{E}(\tau) < \infty$ e $\mathbb{E}(|M_k - M_{k-1}| | \mathcal{F}_{k-1}) \leq C$, pra algum $C > 0$.

prova:
$$M_{\tau \wedge n} = M_0 + \sum_{k=1}^n \mathbb{1}_{\{\tau \geq k\}} (M_k - M_{k-1})$$

② $\tau < \infty \Rightarrow M_{\tau \wedge n} \xrightarrow{n \rightarrow \infty} M_\tau \text{ q.c. } (*)$

$$\begin{aligned} |M_{\tau \wedge n}| &\leq |M_0| + \sum_{k=1}^n \mathbb{1}_{\{\tau \geq k\}} |M_k - M_{k-1}| \\ &\leq |M_0| + C \sum_{k=1}^n \mathbb{1}_{\{\tau \geq k\}} \\ &\leq |M_0| + C \sum_{k=1}^{\infty} \mathbb{1}_{\{\tau \geq k\}} \\ &= |M_0| + C \cdot \tau \quad (**) \end{aligned}$$

$(*) + (**) \xRightarrow{\text{TCD}} \mathbb{E}(M_\tau) = \mathbb{E}(M_0).$

(iv) segue similarmente

Def: $\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq n\} \in \mathcal{F}_n, \forall n \in \mathbb{N}\}$ em geral

• exemplo: $\tau \in \mathcal{F}_\tau$, $X_\tau \in \mathcal{F}_\tau$, $\tau+1 \in \mathcal{F}_\tau$, $\tau-1 \notin \mathcal{F}_\tau$

Prop: τ, σ t.p., $\tau \leq \sigma \Rightarrow \mathcal{F}_\tau \subseteq \mathcal{F}_\sigma$

Teo: $(M_n)_{n \in \mathbb{N}}$ martingal, τ, σ t.p's limitados, $\tau \leq \sigma$
 $\Rightarrow \mathbb{E}(M_\sigma | \mathcal{F}_\tau) = M_\tau$

Cor: $(M_n)_{n \in \mathbb{N}}$ martingal, τ t.p., $\tau \leq N$
 $\Rightarrow \mathbb{E}(M_N | \mathcal{F}_\tau) = M_\tau.$

③ Desigualdades de Doob

Def: Sequência maximal $X_n^* = \max_{0 \leq k \leq n} X_k$

Lem: $(M_n)_{n \in \mathbb{N}}$ submartingal.

$$\Rightarrow \forall n \geq k, \mathbb{E}(M_k \mathbb{1}_A) \leq \mathbb{E}(M_n \mathbb{1}_A), \forall A \in \mathcal{F}_k.$$

prova:

~~Propriedades subMartingal~~

~~$\mathbb{E}(M_n | \mathcal{F}_k) \leq M_k \quad \forall n \geq k$~~

~~$\mathbb{E}(M_n \mathbb{1}_A) \geq \mathbb{E}(M_k \mathbb{1}_A) \quad \forall n \geq k, \forall A \in \mathcal{F}_k$~~

$\bullet M_k \leq \mathbb{E}(M_n | \mathcal{F}_k) \quad \forall n \geq k$

$$\Rightarrow \forall A \in \mathcal{F}_k, M_k \cdot \mathbb{1}_A \leq \mathbb{1}_A \mathbb{E}(M_n | \mathcal{F}_k) = \mathbb{E}(M_n \mathbb{1}_A | \mathcal{F}_k)$$

$$\Rightarrow \mathbb{E}(M_k \mathbb{1}_A) \leq \mathbb{E}(M_n \mathbb{1}_A)$$

Teo: (~~Doob~~ Desigualdade Maximal de Doob):

$(M_n)_{n \in \mathbb{N}}$ submartingal, $M_n \geq 0 \quad \forall n$.

$$\lambda > 0$$

$$\Rightarrow \mathbb{P}(M_n^* \geq \lambda) \leq \frac{\mathbb{E}(M_n \mathbb{1}_{\{M_n^* \geq \lambda\}})}{\lambda} \leq \frac{\mathbb{E}(M_n)}{\lambda}$$

④ Prova: • $\tau := \min\{n \in \mathbb{N} : M_n \geq \lambda\}$

$$\Rightarrow \{\tau > n\} = \{M_n < \lambda, \forall m \leq n\} = \bigcap_{m=1}^n \{M_m < \lambda\} \in \mathcal{F}_n$$

$$\Rightarrow \{\tau \leq n\} = \{\tau > n\}^c \in \mathcal{F}_n$$

$\Rightarrow \tau$ é t.p.

$$\bullet \lambda \mathbb{1}_{\{\tau \leq n\}} \leq M_{\tau \wedge n} \mathbb{1}_{\{\tau \leq n\}} = \sum_{k=0}^n M_k \mathbb{1}_{\{\tau = k\}} \quad (*)$$

$$\bullet \{\tau = k\} \in \mathcal{F}_k \Rightarrow \forall k \leq n, \mathbb{E}(M_k \mathbb{1}_{\{\tau = k\}}) \leq \mathbb{E}(M_n \mathbb{1}_{\{\tau = k\}}) \quad (**)$$

$$\bullet \{M_n^* \geq \lambda\} = \left\{ \max_{m \leq n} M_m \geq \lambda \right\} = \{\tau \leq n\} \quad \begin{matrix} (*) \\ ** \end{matrix}$$

$$\bullet \lambda \mathbb{P}(M_n^* \geq \lambda) \stackrel{(**)}{=} \lambda \mathbb{P}(\tau \leq n) = \mathbb{E}(\lambda \mathbb{1}_{\{\tau \leq n\}})$$

$$\stackrel{(*)}{\leq} \mathbb{E}\left(\sum_{k=0}^n M_k \mathbb{1}_{\{\tau = k\}}\right) \stackrel{**}{=} \sum_{k=0}^n \mathbb{E}(M_k \mathbb{1}_{\{\tau = k\}})$$

$$\leq \sum_{k=0}^n \mathbb{E}(M_n \mathbb{1}_{\{\tau = k\}}) = \mathbb{E}\left(M_n \sum_{k=0}^n \mathbb{1}_{\{\tau = k\}}\right)$$

$$\stackrel{(**)}{=} \mathbb{E}(M_n \mathbb{1}_{\{\tau \leq n\}}) \stackrel{(*)}{=} \mathbb{E}(M_n \mathbb{1}_{\{M_n^* \geq \lambda\}})$$

Cor: $p \geq 1 \Rightarrow (|M_n|^p)_{n \geq 0}$ é submartingal não-negativo (se $\mathbb{E}|M_n|^p < \infty$)

$$\Rightarrow \mathbb{P}(M_n^* \geq \lambda) \leq \frac{\mathbb{E}(|M_n|^p)}{\lambda^p}$$

⑤ $p > 1$
Lemma: $X, Y \geq 0$ t.q. $Y \in L^p$ e $\lambda P(X \geq \lambda) \leq E(Y \mathbb{1}_{\{X \geq \lambda\}})$
 $\forall \lambda \geq 0$

$$\Rightarrow \|X\|_p \leq \frac{p}{p-1} \|Y\|_p \quad (\text{em particular, } X \in L^p).$$

Prova: • $X_n = X \wedge n$.

• $X_n \rightarrow X$ q.c

$$\text{Fatou} \Rightarrow \|X\|_p \leq \liminf_{n \rightarrow \infty} \|X_n\|_p$$

$$\Rightarrow \text{Basta provar } \|X_n\|_p \leq \frac{p}{p-1} \|Y\|_p.$$

$$\bullet \mathbb{1}_{\{X_n \geq \lambda\}} = \mathbb{1}_{\{X \geq \lambda\}} \mathbb{1}_{\{n \geq \lambda\}}$$

$$\Rightarrow E(Y \mathbb{1}_{\{X_n \geq \lambda\}}) = E(Y \mathbb{1}_{\{X \geq \lambda\}}) \mathbb{1}_{\{n \geq \lambda\}}$$

$$\geq \lambda P(X \geq \lambda) \mathbb{1}_{\{n \geq \lambda\}} = \lambda E(\mathbb{1}_{\{X \geq \lambda\}}) \mathbb{1}_{\{n \geq \lambda\}}$$

$$= \lambda E(\mathbb{1}_{\{X \geq \lambda\}} \mathbb{1}_{\{n \geq \lambda\}}) = \lambda P(X_n \geq \lambda) \quad \forall \lambda \geq 0$$

$$\bullet Z^p = p \int_0^Z t^{p-1} dt = p \int_0^\infty t^{p-1} \mathbb{1}_{\{t \leq Z\}} dt$$

$$\text{Fubini} \Rightarrow E X_n^p = p \int_0^\infty t^{p-1} P(X_n \geq t) dt$$

Tonelli

$\Rightarrow$

$$⑥ \quad \mathbb{E}(X_n^p) \leq p \int_0^\infty t^{p-2} \mathbb{E}(Y \mathbb{1}_{\{X_n \geq t\}}) dt$$

$$= p \mathbb{E}\left(Y \int_0^\infty t^{p-2} \mathbb{1}_{\{X_n \geq t\}} dt\right)$$

$$= p \mathbb{E}\left(Y \int_0^{X_n} t^{p-2} dt\right) = p \mathbb{E}\left(Y \frac{X_n^{p-1}}{p-1}\right) = \frac{p}{p-1} \langle Y, X^{p-1} \rangle$$

Escolha q t.q. $\frac{1}{p} + \frac{1}{q} = 1$.

$$\text{Holder} \Rightarrow \|X_n\|_p^p \leq \frac{p}{p-1} \|Y\|_p \underbrace{\|X_n^{p-1}\|_q}_{= \|X_n\|_p^{p-1}}$$

$$\Rightarrow \|X_n\|_p \leq \frac{p}{p-1} \|Y\|_p$$

Teo (Desigualdade L^p de Doob): $(M_n)_{n \in \mathbb{N}}$ submartingal $\tilde{\pi}$ -negativo
 $\Rightarrow \forall p > 1, \forall n \geq 0, \|M_n^*\|_p \leq \frac{p}{p-1} \|M_n\|_p$.

prova $M_n^* \in L^p \Leftrightarrow M_n \in L^p \dots$

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Desigualdade de Doob para martingais L^1 .

$$\begin{aligned} \bullet \mathbb{E}(X_n) &= \int_0^\infty \mathbb{P}(X_n \geq t) dt \leq 1 + \int_1^\infty \mathbb{P}(X_n \geq t) dt \\ &\leq 1 + \int_1^\infty \frac{1}{t} \mathbb{E}(Y \mathbb{1}_{\{X_n \geq t\}}) dt = 1 + \mathbb{E}\left(Y \int_1^\infty \frac{1}{t} \mathbb{1}_{\{X_n \geq t\}} dt\right) \end{aligned}$$

$$\bullet \log^+ x := \max\{\log x, 0\} = \int_1^\infty \frac{1}{t} \mathbb{1}_{\{x \geq t\}} dt$$

$$\Rightarrow \mathbb{E}(X_n) \leq 1 + \mathbb{E}(Y \log^+ X_n) \leq 1 + \mathbb{E}\left(Y \log^+ Y + \frac{X_n}{e}\right)$$

$$\bullet \forall a, b > 0, \quad a \log^+ b \leq a \log^+ a + \frac{b}{e}$$

$$\Rightarrow \mathbb{E}(X_n) \leq \frac{e}{e-1} \left(1 + \mathbb{E}(Y \log^+ Y)\right)$$

em particular, quando $X_n = M_n^*$ e $Y = M_n$,

$$\mathbb{E}(M_n^*) \leq \frac{e}{e-1} \left(1 + \mathbb{E}(M_n \log^+ M_n)\right) \quad \forall n \in \mathbb{N}. \quad \blacksquare$$